Common net issue theory

1. **List 5 common hardware issues that can cause network problems in a typical office setup, and for each, describe one symptom that helps identify it.**

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1. Faulty Network Cable

Symptom: The computer shows “No Internet” or the network connection keeps disconnecting.

2. Damaged Network Port

Symptom: The device does not detect the Ethernet cable even when the cable is properly connected.

3. Switch Failure

Symptom: Multiple computers connected to the same switch suddenly lose network connectivity.

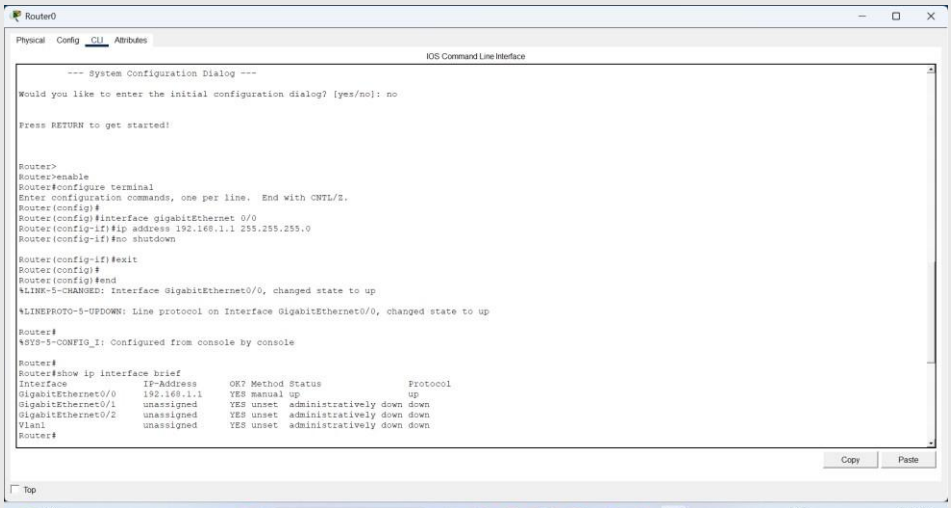
4. Faulty Network Interface Card (NIC)

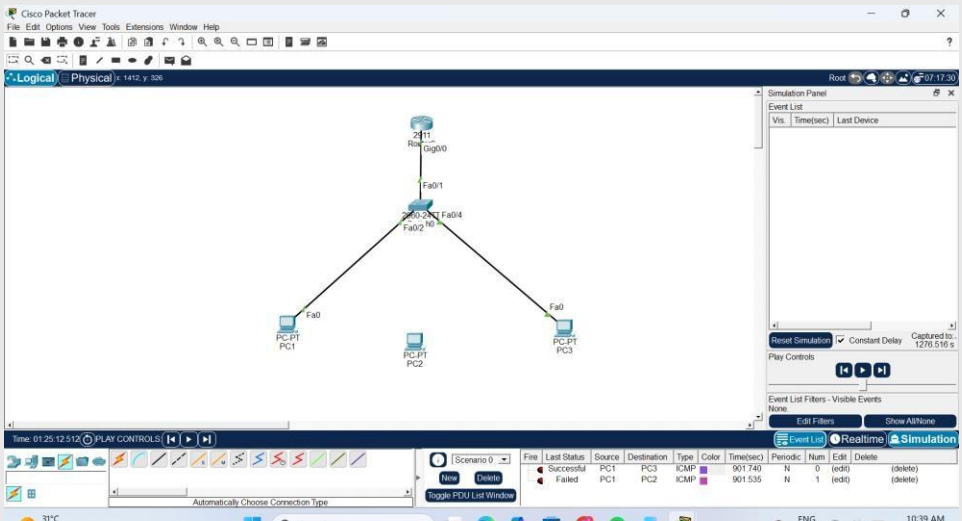
Symptom: One computer cannot connect to the network, while other computers work normally.

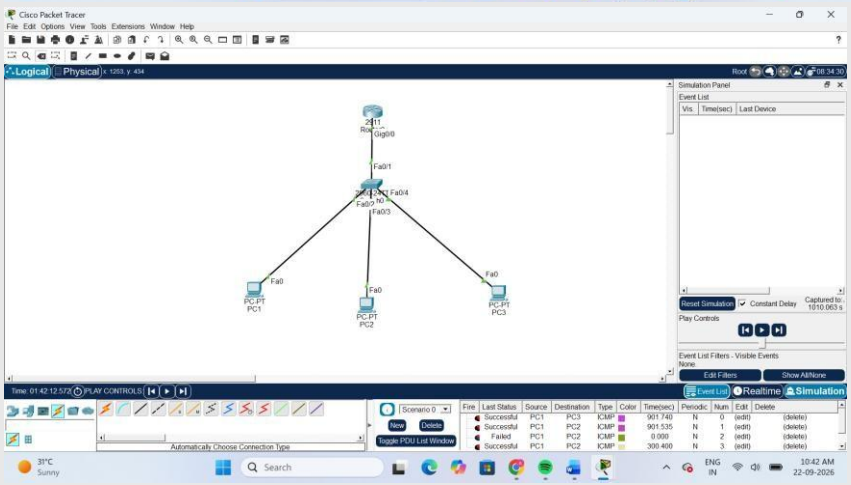
5. Router Hardware Failure

Symptom: Devices connected to the router cannot access the network or Internet, and the router may show abnormal or no status lights.

2.Open Cisco Packet Tracer or any network simulator, set up a small network with 3 PCs, 1 switch, and 1 router, and intentionally disconnect one device. Use the simulator's troubleshooting tools to identify which device is faulty and document the steps you took.







Troubleshooting Steps

1. I created a small network using 3 PCs, 1 switch, and 1 router in Cisco Packet Tracer.
2. I configured the IP addresses of all three PCs and the router.
3. I first tested the network using a Simple PDU (ICMP).
4. The connection between PC1 and PC3 was successful, confirming that the network was working properly.
5. I intentionally disconnected PC2 from the switch to create a network fault.
6. I sent an ICMP packet from PC1 to PC2, which resulted in a Failed status.
7. I checked PC2's IP configuration and found it was correctly configured.
8. I then checked the physical connection and identified that PC2 was disconnected from the switch.
9. I reconnected PC2 to Switch Fa0/3 using a Copper Straight-Through cable.
10. I tested the connection again using Simple PDU.
11. The PC1 → PC2 connection was successfully restored.

Fault Identified

Fault: PC2 was physically disconnected from the switch.

Symptom: ICMP communication from PC1 to PC2 failed.

Solution: Reconnected PC2's FastEthernet0 interface to Switch Fa0/

**3.Create a troubleshooting checklist for diagnosing slow internet issues in a home WiFi network, including at least 8 steps covering both hardware and software checks.

Hint: Think about**

checking cables, router status lights, device settings, and running ping or speed tests.

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Slow Internet Troubleshooting Checklist

1. **Check the cables**
   Make sure the router's power and Ethernet cables are properly connected and not damaged.

2. **Check router status lights**
   Check the Power, Internet/WAN, and Wi-Fi lights. Make sure they are showing normal status.

3. **Restart the router**
   Turn off the router, wait for about 30 seconds, and turn it on again. Wait for the connection to come back.

4. **Check Wi-Fi signal strength**
   Move closer to the router and check whether the internet becomes faster. A weak signal can cause slow speeds.

5. **Check connected devices**
   Disconnect unnecessary devices because multiple devices using the same Wi-Fi can reduce available bandwidth.

6. **Check device Wi-Fi settings**
   Make sure Wi-Fi is enabled and the device is connected to the correct network. Forget and reconnect to the Wi-Fi if needed.

7. **Run a ping test**
   Open Command Prompt and run ping 8.8.8.8. High response times or packet loss can indicate a network problem.

8. **Run a speed test**
   Use a reliable internet speed test and compare the download and upload speeds with the speed promised by the ISP.

9. **Check for background usage**
   Pause large downloads, cloud synchronization, Windows updates, or streaming applications that may be consuming bandwidth.

10. **Test with another device**

    Connect another phone or laptop to the same Wi-Fi. If all devices are slow, the problem may be with the router or ISP. If only one device is slow, check that device's settings or software.

4.You receive a complaint that a user cannot access Instagram and YouTube on their laptop, but WhatsApp Web works fine. Write down the first 5 troubleshooting steps you would perform to identify if the issue is with hardware, software, or network configuration.

